Haoyu Sheng

PhD · Economics

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Education

Brown University *Providence, RI*

Ph.D in Economics *2020 - 2025*

M.Sc in Computer Science *2022 - 2025*

- Research Areas: Macroeconomics, Production Networks, Finance, Machine Learning.

Williams College *Williamstown, MA*

B.A. in Computer Science and Economics with Highest Honors *2016 - 2020*

Working Papers

"Measuring Innovation Through the Commercialization of New Ideas"

Abstract: Innovation extends beyond patents and R&D. I propose a holistic measure of innovation using text data from startup websites and public company 10-K filings. I designate successful young startups as the innovation benchmark and measure how much public companies' business descriptions evolve toward the benchmark over time. My measure captures firms' adoption of emerging technologies, pivots into new markets, and imitation of successful entrants. It correlates with R&D and patent-based innovation proxies and is available for firms that do not engage in patenting or R&D. Applying this measure, I find that younger firms, those in less concentrated markets, struggling firms, and more liquid firms are more likely to innovate. I also find that innovation improves future financial performance: a one standard deviation increase in innovation raises average excess returns by 2 percentage points and earnings normalized by sales by 0.4 percentage points over the next five years.

"Unemployment in a Production Network," with Finn Schüle

Abstract: We model a production network economy with sectoral and occupational unemployment by incorporating matching between job seekers across various occupations and employers in different production sectors. We derive expressions that unpack how the impact of microeconomic shocks on output and unemployment depends on the interaction between the network linkages, search costs, and changes in labor market tightness. When labor markets are slack, our model predicts larger output and employment responses because the network-adjusted labor productivity gain outweighs search costs. Calibrating our model to the U.S. economy, we demonstrate that our model significantly amplifies the response of aggregate output and unemployment to productivity shocks in any sector and changes the relative importance of sectors to aggregate output and unemployment. Our model nearly doubles the output response compared to an efficient production network and triples the unemployment response compared to a multi-sector search model following a productivity shock to durable manufacturing.

"Gender Differences in Economic Seminars," with Pascaline Dupas, Amy Handlan, Alicia Sasser Modestino, Muriel Niederle, Mateo Seré, Justin Wolfers, and the Seminar Collective, *Revise and resubmit* at the *American Economic Review*

Abstract: We assess whether men and women are treated differently when presenting their research in economics seminars. We collected data on every interaction between presenters and audience members across thousands of seminars, job market talks and conference presentations, leveraging both human judgment and audio processing algorithms to measure the number, tone and type of interruptions. Within a seminar series, women are interrupted more than men, and this finding holds when controlling for characteristics of the presenter and their paper topic. This differential treatment appears to reflect in part greater engagement with female speakers—resulting in larger attendance and seminar engagement, and in part greater hostility toward female speakers—resulting in more negative, mid-sentence, and declarative interruptions.

"Gender and Tone in Recorded Economics Presentations: Audio Analysis with Machine Learning," with Amy Handlan

Abstract: This paper measures seminar dynamics using a replicable, scalable, machine-learning approach and finds a gender-tone gap in economics presentations. We train a deep convolutional neural network to impute labels for gender and tone-of-voice. We apply this to recorded presentations from the 2022 NBER Summer Institute to measure tone at a high frequency, which allows us to provide novel results on how economists interact with each other in talks. We find that female economists are less likely to be spoken to in a positive tone and more likely to be addressed with a serious and stern tone. Female economists are also more likely to speak in a positive tone. Overall, we show that gender differences in economics presentations exist across fields and presentation formats.

Published Papers

"When Rooks Miss: Probability through Chess," with Steven Miller and Daniel Turek
The College Mathematics Journal, 2021

"Assessing Post-hoc Explainability of the BKT Algorithm," with Iris Howley and Tongyu Zhou
2020 AAAI/ACM Conference on AI, Ethics, and Society (AIES'20), 2020

"Knowledge Distillation for Recurrent Neural Network Language Modeling with Trust Regularization," with Yangyang Shi, Mei-Yuh Hwang, and Xin Lei

ICASSP 2019 - 2019 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), 2019

Teaching Experiences

Fall 2024	Investments II, Professor Kwon, Brown University
Spring 2022	Machine Learning, Text Analysis, and Economics, Professor Handlan, Brown University
Spring 2020	Game Theory, Professor Rai, Williams College
Fall 2019	Global Macro Instability and Finance, Professor Phelan, Williams College
Fall 2017, Spring 2019	Intermediate Microeconomics, Professors Rai and Sheppard, Williams College

Honors and Awards

2023	Bravo Center Research Award, Brown University
2022 - 2024	Open Graduate Education Fellowship, Brown University
2020	Graduate Fellowship, Brown University
2019	Carl Van Duyne Prize in Economics , Williams College
2018	Class of 1960 Scholar in Computer Science, Williams College
2018	Class of 1960 Scholar in Economics, Williams College

Presentations and Workshops

2024	Machine Learning in Economics Summer Institute (Chicago Booth), Brown Macro Breakfast
2023	ASSA CSWEP, Brown Macro Breakfast, WEAI, Midwest Macroeconomics Meetings
2022	Brown Macro Breakfast, NBER Behavioral Macro Bootcamp, NBER Heterogeneous-Agent Macro Workshop

Other Experiences

Summer 2024	Quantitative Research, Susquehanna International Group
Summer 2018	Deep Learning Research, Mobvoi AI Lab
Summer 2017	Quantitative Research, Hutchin Hill Capital

Skills

Programming Languages Python (pandas, numpy, nltk, tensorflow, pytorch, and librosa), Java, Javascript, R, C++